Fiscal Year:	FY 2007	**Task Last Updated:**	FY 03/04/2008
PI Name:	LiCata, Vince J Ph.D.		
Project Title:	The Effects of Microgravity on Protein-Ligand Interactions		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Molecular structures and physical interactions		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2000 Space Biology 00-OBPR-01
Start Date:	08/01/2001	**End Date:**	07/31/2007
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 07/31/2007 per A. Chu (2/07) NOTE: received NCE until 2/28/2007, per A. Chu (9/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Performance Goal No.:			
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Task Description:	In this proposal, several specific chemical kinetics and protein-ligand systems will be examined in parallel in ground based versus reduced gravity experiments: 1) the association of adipocyte lipid binding protein (ALBP) with lipids; 2) the association of bovine serum albumin and lipids; and 3) the association of DNA polymerase 1 (Pol 1) with DNA; and the reaction of ascorbic acid and DCIP as a function of pH. Parallel kinetic and equilibrium binding assays will be carried out in all three systems in the laboratory and in reduced gravity during parabolic flights on NASA's KC-135A aircraft. Assays will also be performed as a function of viscosity to quantitate the gravitational effect on the degree of diffusion control of each interaction.. Because of the established relationship between electrostatic attraction and the rates of diffusion of charged substrates toward enzymes, the effects of altering the electrostatics of the protein-ligand attractions in these systems will also be examined.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This proposal examines the effects of microgravity on protein-ligand interactions, with a focus on fast and diffusion controlled reactions. Each of the biochemical reactions to be studied is physiologically important: 1) adipocyte lipid binding protein (ALBP) is linked to both obesity and the development of Type II diabetes; 2) serum albumin is responsible for transport of fatty acids and other lipids in the bloodstream; 3) Pol 1 is involved in DNA replication and repair in vivo, and is a model system for understanding DNA polymerases in general. Microgravity effects on any of these reactions would have important physiological consequences. The main focus of this proposal, however, is to utilize these systems as benchmark reactions to test the effects of microgravity on protein-ligand interactions in general. Microgravity, with it's reduced convective mixing and lack of molecular buoyancy or sedimentation, is clearly likely to alter the rates, and possibly the equilibria of protein-ligand interactions, especially of fast and diffusion limited reactions. Even relatively small changes in these equilibria or rates could have significant physiological effects. Gravitational effects may be wholly or partially responsible for the fact that even the fastest known enzymes never attain the theoretical limit for diffusion controlled reactions.
Task Progress:	GOAL: Overall goal: to determine if microgravity alters protein-substrate binding kinetics and equilibria. SUMMARY OF ACCOMPLISHMENTS: I) Initial background work on salt dependence of the kinetics and lipid binding equilibria for the adipocyte lipid binding protein was described in detail in the initial publication appeared in 2003: Schoeffler, A.J., Ruiz, C.R., Joubert, A.M., Yang, X., and LiCata, V.J., 2003, Salt modulates the stability and lipid binding affinity of the adipocyte lipid binding protein, J. Biol. Chem. 278, 33268-33275. Two other publications are in preparation from this part of the project: Yang, X., Ruiz, C.R., and LiCata, V.J. Ion specific effects on lipid binding by the adipocyte lipid binding protein, in preparation. Schoeffler, A.J., Ruiz, C.R., Liu, C.-C., Bernlohr, DA., and LiCata, V.J., Electrostatic steering during lipid binding to the adipocyte lipid binding protein as reveled in a charge-deletion mutant, in preparation. II) The primary results of the project were obtained over the course of 5 sets of flights on NASA's Reduced Gravity Office aircraft. 1st flight week: 8/31/04 - 9/2/04 2nd flight week: 5/8/06 -5/12/06 3rd flight week: 3/20/07-3/23/07 4th flight week: 5/22/07 - 5/24/07 + 6/19/07 (weather delay makeup flight) 5th flight week: 7/30/07 - 8/03/07 Key flight experiment results are described below. The DCIP chemical reaction was designed as the test/standardization reaction for the onboard equipment, and was not expected to show alteration in microgravity. Results confirm that the equipment designed for onboard flights worked, and that as predicted, the DCIP reaction was unaffected by gravity. After confirmation of the DCIP standard reaction on the first set of flights, protein-ligand association studies were commenced. On the second set of flights, the association rate kinetics for the binding of the fluorescent lipid ANS to the adipocyte-lipid binding protein was examined. Results suggested a microgravity-induced change in reaction kinetics for this system, but the high experimental noise level for data collected in-flight did not allow such an alteration to be statistically verified for this system. Similar results were obtained for the association of bovine serum albumin (BSA) and ANS on these flights. On the next set of flights, we began to assay enzyme kinetics rather than direct substrate binding kinetics. The first system examined was alkaline phosphatase. All reactions were measured both during the microgravity portion of each parabola, and during the 1.7g portion of each parabola. A full series of kinetic reactions for alkaline phosphatase was collected at both zero g and at 1.8g at 0%, 10%, 20%, and 30% glycerol. On-board data were significantly less precise than on-ground data (significantly higher random noise). Apparent differences between 0g and 1g are strongly suggested by the data, where the viscosity dependence for alkaline phosphatase is quite different in 0g versus 1g. However, again, the in-flight data precision problem, and resultant low statistical significance of the differences required additional data collection on this system. In flight sets 4 and 5, we primarily examined the kinetics of "fast" and "slow" forms of the enzyme acetylcholinesterase (AChE). These flight sets involved a further experimental data collection redesign to decrease the high experimental noise characteristic of in-flight data. On flight sets 4 and 5, data were collected continuously as the aircraft progressed

	through the parabolic flight pattern. Data were then separated into 0g sequences and 2g sequences post-flight. The data indicate a clear change in the enzyme kinetics for the slow form of AChE in microgravity versus macrogravity. The kinetic parameters in microgravity indicate that the reaction rate for slow acetylcholinesterase is enhanced in microgravity. Section II Results: The primary goals were to determine if microgravity alters the reaction rates for biological reactions. Small apparent effects of microgravity have been observed in several of the systems measured during this study; however, the harsh measurement conditions during flight consistently resulted in high noise-to-signal ratios for in-flight data. Continued redesign of the experimental equipment improved data quality to the point where microgravity effects on at least one enzyme system (slow form acetylcholinesterase) has been conclusively demonstrated. The demonstration that not all enzymatic systems are altered by microgravity is also an important result of this study. A preliminary summary account of the section II results are described in: LiCata, V.J., Enzyme Kinetics in Microgravity, in Risk and Exploration: The Earth as a Classroom, Louisiana State University Press, in press. A detailed report of these results is currently in preparation: Liu, C.-C., Thompson, G.S., Bell, J.C., and LiCata, V.J., Effects of Microgravity on Enzymatic Reactions, in preparation.
Bibliography Type:	Description: (Last Updated: 01/31/2008)
Articles in Peer-reviewed Journals	Schoeffler AJ, Ruiz CR, Joubert AM, Yang X, LiCata VJ. "Salt modulates the stability and lipid binding affinity of the adipocyte lipid-binding proteins." J Biol Chem. 2003 Aug 29;278(35):33268-75. PMID: 12794068 , Aug-2003
Papers from Meeting Proceedings	LiCata VJ. "Enzyme Kinetics in Microgravity." Risk and Evaluation II: The Earth as a Classroom, Louisiana State University, Baton Rouge, LA, October 28-30, 2007. In: Risk and Evaluation II: The Earth as a Classroom, in press, 2007. , Dec-2007